

COMPUTER ENGINEER • FULL-STACK DEVELOPER

# Ms. Natthamon Kosem

Full-Stack Software Engineer | B.Eng Computer | 4+ Years Exp | Bangkok, TH

**Phone:** 099-564-9387 **Email:** nutkosem@gmail.com **Location:** Nonthaburi / Bangkok, Thailand

## PERSONAL PROFILE

Computer engineer and full-stack software engineer with **4+ years of professional experience** across the entire product lifecycle.

Proven capability from reactive web frontends (Vue/Nuxt, React, TypeScript) to mission-critical backend microservices, RESTful APIs, and enterprise relational databases (**MySQL & Oracle DB**).

Experienced across all testing levels (unit, functional, integration, system, UAT) with automated CI/CD pipelines and industrial IoT integration.

## AREAS OF EXPERTISE

**CORE LANGUAGES**

Python | JavaScript (ES6+) | TypeScript | PHP | SQL | C / C++ | HTML5 / CSS3

**BACKEND & APIS**

RESTful API Architecture | Node.js (Express) | PHP (Laravel MVC) | Microservices | OpenAPI / Swagger Specs | Message Queues & Workers

**DATABASES & PERSISTENCE**

MySQL | Oracle Database | PL/SQL Stored Procedures | Schema Normalization | Query Optimization & Indexing | Transaction Integrity

**FRONTEND & WEB**

Vue.js / Nuxt 4 | React & Redux | Tailwind CSS | React Hook Form | jQuery / DOM | Responsive Design

**DEVOPS, CLOUD & TESTING**

GitLab CI/CD Pipelines | Git / GitHub | Docker Containerization | Postman Automated Testing | Unit & Functional Testing | Linux (Bash CLI) | Vercel Serverless

## CORE STRENGTHS

- End-to-end full-stack web & API development
- Database modeling, complex queries & optimization
- Rigorous testing across all SDLC phases
- Industrial IoT & hardware protocol integration
- Fast adoption of new tools & attention to detail
- Strong analytical problem solving & team collaboration

## WORK EXPERIENCE

### Rudy Technology — 2022 – Present

**Full-Stack Developer (Bangkok, Thailand)**

- Built and maintained **RESTful APIs** connecting web applications and mobile clients with central backend services.
- Designed relational database schemas in **MySQL & Oracle DB**, writing optimized SQL queries, stored procedures, and MVC controllers.
- Developed responsive web interfaces and user dashboards, translating wireframes into functional frontend components.
- Integrated low-code mobile CRM applications with backend databases and custom API endpoints.
- Configured and managed **GitLab CI/CD** pipelines for automated testing and server deployments.
- Authored OpenAPI/Swagger documentation to ensure clear API contracts between frontend and backend teams.

### Freelance Industrial Automation & HMI — 2023 – 2024

**Industrial Automation Engineer (Independent / Part-Time Project)**

- Developed the **VBX_APUST** HMI control software for automated dispensing production lines.
- Interfaced hardware across multiple protocols: industrial PLC via **Modbus TCP** (16 DI / 16 DO), Keyence barcode scanners (TCP), and Cognex vision cameras (HTTP REST).
- Built a deterministic state machine workflow ensuring operator safety and reliable robot dispensing.

### So-at Solution — 2021

**Software Engineer Intern (Co-operative Training)**

- Investigated and resolved web service issues, troubleshooting database locks and request timeouts.
- Developed backend MVC controllers and SQL functions for internal web systems.
- Created automated **Postman test collections** to validate API endpoints between staging and production.
- Collaborated with UI/UX designers to build clean, responsive web pages for enterprise users.

### SCG (Siam Cement Group), Bang Sue — 2020

**Engineering Intern (Student Training)**

- Researched enterprise database structures and analyzed query execution plans to improve retrieval performance.
- Built an end-to-end IoT prototype combining microcontroller hardware, sensors, and backend data logging.
- Designed user-friendly interfaces for on-site field engineers to monitor factory equipment.

## FEATURED PROJECTS

**3rd-Party Mobile API Integration Gateway** — 2023

High-throughput API proxy layer orchestrating third-party vendor APIs with caching, token rotation, and rate-limiting.

Node.js • RESTful API • Microservices • Security

**Unified Core RESTful API Platform** — 2022

Core backend service endpoints with robust auth, transaction handling, schema validation, and Postman test automation.

Node.js • Laravel • MySQL • Postman • CI/CD

**AI Visual Similarity & NLP Pipelines** — 2020–2023

Computer vision cosine similarity verification (OpenCV) and NLP intent classification pipelines in Python.

Python • OpenCV • NLP • Text Analytics

**Web CRM Platform (React & Redux)** — 2021–2022

Interactive business CRM application built with React, Redux, React Hook Form, and backend SQL database integration.

React • Redux • TypeScript • CSS • SQL API

## EDUCATION HISTORY

### SRIPATUM UNIVERSITY — Graduated: 2021

**Bachelor of Computer Engineering (B.Eng) • GPA: 3.27**

- **President**, Computer and Technology Organization • **Vice President**, Student Council (Samo Sripatum)
- **National Award:** Distinguished IT Student Award for Ethics & Morality (InCIT / nCIT 2020)